



ADRAM
TECHNOLOGIES · LEARNING HUB

LEARNING NOTES · FROM ZERO TO HERO

Networking

From your first cable to designing secure business networks: a step-by-step path to becoming a network technician and engineer.

2

levels

6

topics

24

notes

50 minutes

of reading

ADRAM Technologies · localhost:5173/learning/networking
Edition of October 2026. The latest version is always online.

ABOUT THIS FIELD

Networking

This path is for **complete beginners** who want to understand how networks work, and for anyone preparing for a career in IT support or network engineering.

- **No experience needed.** The Zero level starts with what a network is.
- **Hands-on from the start.** Labs use the computer you already have, and later the free Cisco Packet Tracer simulator.
- **Work at your own pace.** Follow the levels in order and tick each note off as you finish it.

By the end of the Foundations level you will be able to explain how devices find each other on a network, read and plan IP addresses, and troubleshoot common connection problems.

HOW THE LEARNING PATH WORKS

Every field runs through five levels. Start at the first, follow the notes in order, and tick each one off as you finish it.

1 Zero

No experience needed: the very first ideas, explained simply.

2 Foundations

The core concepts every professional uses every day.

3 Intermediate

Configure, build and troubleshoot real set-ups. (notes coming soon)

4 Advanced

Design, secure and scale with confidence. (notes coming soon)

5 Hero

Expert topics, research and certification-level depth. (notes coming soon)

CONTENTS

What is inside

Level 1: Zero 5

What is a computer network?	6
What is a computer network?	6
Devices that make a network work	8
Cables, fibre and Wi-Fi: how data travels	10
Lab: explore the network you are on	12
Types of networks	15
LAN, WAN, Wi-Fi and the internet: types of networks	15
Networking words every beginner should know	17

Level 2: Foundations 19

IP addressing & subnetting	20
What is an IP address?	20
Binary for networking	23
Subnetting step by step	25
Subnetting cheat sheet	27
Lab: subnet a small office network	29
IPv6 essentials	31
The OSI and TCP/IP models	33
The OSI model: seven layers explained	33
The TCP/IP model and encapsulation	35
TCP and UDP: reliable or fast	37
Common ports and protocols	39
Ethernet and switching basics	41
Ethernet frames and MAC addresses	41
How a switch learns and forwards	43
ARP: finding a MAC address from an IP address	45
Lab: build your first network in Packet Tracer	47

Essential network services	49
DHCP: automatic IP addressing	49
DNS: turning names into addresses	51
NAT and PAT: sharing one public address	53
Lab: watch DHCP and DNS at work	55

LEVEL 1 OF 5

Zero

No experience needed: the very first ideas, explained simply.

What is a computer network?

The very first ideas: what a network is, what it is made of and what it lets us do.

- What is a computer network?
- Devices that make a network work
- Cables, fibre and Wi-Fi: how data travels
- Lab: explore the network you are on

Types of networks

From the network in your pocket to the internet itself.

- LAN, WAN, Wi-Fi and the internet: types of networks
- Networking words every beginner should know

ZERO · WHAT IS A COMPUTER NETWORK?

NOTE · 2 min read

What is a computer network?

A simple introduction: what a network is, what it lets us share and the four parts every network has.

WHAT YOU WILL LEARN

- Explain in your own words what a computer network is
- Give examples of what networks let us share
- Name the four parts every network has
- Describe the difference between a client and a server

A simple idea

A **computer network** is two or more devices connected so that they can exchange information and share resources.

You use networks all day without noticing. When you send a WhatsApp message, print a document at the office, pay with mobile money or watch a video, information travels across one or more networks to reach the right place.

What networks let us share

- **Information:** files, messages, photos, videos and records.
- **Resources:** one printer or one internet connection shared by a whole office.
- **Services:** email, websites, online banking and cloud applications.

Before networks, sharing a file meant copying it onto a disk and carrying it to the next computer. Networks made sharing instant.

The four parts of every network

Part	What it does	Examples
End devices	Create information or use it	Laptops, phones, printers, servers
Network devices	Move information to the right place	Switches, routers, Wi-Fi access points
Media	Carry the signals	Copper cables, fibre-optic cables, radio waves (Wi-Fi)
Protocols	The rules devices follow to understand each other	TCP/IP, HTTP, Wi-Fi standards

NOTE

A protocol is like a shared language with agreed manners. Two devices can only communicate if they follow the same protocols.

Clients and servers

Most communication on a network follows a simple pattern:

- A **client** asks for something. Your phone's browser asking for a web page is a client.
- A **server** answers. The computer that stores the website and sends it back is a server.

The same device can be both: your laptop is a client when it opens a website and a server when it shares a folder with a colleague.

Why learn networking?

Every organisation, from a small shop in Freetown to a bank or a hospital, depends on its network. When the network stops, work stops. People who can build, secure and fix networks are needed everywhere.

TIP

As you go through this path, look around you: count the networked devices in your home or office. You will be surprised how many there are.

Key points

- A network connects devices so they can share information, resources and services.
- Every network has end devices, network devices, media and protocols.
- Clients ask for services; servers provide them.

NOTE · 2 min read

Devices that make a network work

End devices and the network devices that connect them: switches, routers, access points, modems, firewalls and servers.

WHAT YOU WILL LEARN

- Tell end devices and network devices apart
- Explain what a switch, a router and an access point do
- Describe the job of a modem and a firewall
- Recognise the devices inside a typical home or office router

End devices

End devices are where information starts or ends. You use them directly.

- Desktop computers and laptops
- Phones and tablets
- Printers and scanners
- Servers
- Security cameras, smart TVs and point-of-sale machines

Each end device needs a way to connect to the network, called a **network interface**: an Ethernet port for a cable, a Wi-Fi adapter for wireless, or both.

Network devices

Network devices sit between end devices and move information to the right place.

Device	What it does	Think of it as
Switch	Connects devices within the same network and sends each message only to the device it is meant for	The post room inside one office building
Router	Connects different networks together, for example your office network to the internet	The post office that sends letters to other towns
Wireless access point	Lets devices join the network over Wi-Fi	A door into the network without a cable
Modem / ONT	Connects your network to your internet provider's line (telephone, cable or fibre)	The translator between your network and your provider
Firewall	Checks traffic coming in and going out and blocks what is not allowed	The security guard at the gate

Device	What it does	Think of it as
Server	Provides services to other devices: files, email, websites, applications	The shop that serves customers

NOTE

Switches work with **MAC addresses** (the hardware address of each network card). Routers work with **IP addresses** (the logical address of each device). You will meet both in the Foundations level.

The box on your desk does several jobs

The "router" an internet provider installs in a home or small office is usually several devices in one box:

1. A **modem** to connect to the provider
2. A **router** to connect your network to the internet
3. A small **switch** with a few Ethernet ports
4. A **wireless access point** for Wi-Fi
5. A basic **firewall**

In larger organisations these are separate devices, so each can be more powerful and managed on its own.

TIP

Look at the back of your home or office router. The ports labelled LAN (usually yellow) are the switch; the port labelled WAN or Internet connects to the provider.

Key points

- End devices create or use information; network devices move it.
- A switch connects devices in the same network; a router connects different networks.
- Access points add Wi-Fi, modems connect to the provider and firewalls control what gets in and out.

NOTE · 3 min read

Cables, fibre and Wi-Fi: how data travels

The three kinds of network media, their speeds and limits, and the difference between bits and bytes.

WHAT YOU WILL LEARN

- Compare copper cable, fibre-optic cable and Wi-Fi
- Know the 100-metre limit of copper Ethernet cables
- Explain the difference between bandwidth and throughput
- Convert between bits and bytes

Three ways to carry data

Network **media** are what carry the signals between devices. There are three main kinds.

Medium	How it carries data	Typical use	Strengths	Limits
Copper (twisted pair)	Electrical signals	Connecting computers, printers and cameras to switches	Cheap, easy to install, powers devices (PoE)	100 m per cable run, affected by electrical interference
Fibre optic	Pulses of light	Links between buildings, to the internet provider, between switches	Very long distances, very fast, not affected by electrical interference	More expensive, needs care and special tools
Wireless (Wi-Fi)	Radio waves	Laptops, phones and tablets	No cables, people can move around	Slower and less reliable than cable, walls weaken the signal, must be secured

Copper Ethernet cables

The cable you see in most offices is **unshielded twisted pair (UTP)** with an **RJ45** connector at each end. The pairs of wires are twisted to reduce interference.

Category	Common speed	Notes
Cat 5e	Up to 1 Gbps	Still common in older buildings
Cat 6	1 Gbps (10 Gbps over short runs)	A good choice for new offices
Cat 6A	10 Gbps	For demanding networks and future growth

WARNING

A single copper Ethernet run should be no longer than **100 metres**, including the patch cables at each end. Longer runs cause errors and slow, unreliable connections.

Fibre-optic cables

Fibre carries light through very thin strands of glass.

- **Multi-mode fibre** is used for shorter distances, such as inside a building or a campus.
- **Single-mode fibre** carries light over many kilometres and is used by internet providers and between sites.

Wi-Fi

Wi-Fi uses radio waves in two main bands:

- **2.4 GHz** reaches further and passes through walls better, but is slower and more crowded.
- **5 GHz** is faster with less interference, but has a shorter range.

Newer standards such as **Wi-Fi 6 (802.11ax)** handle many devices at once better than older ones.

Speed: bits, bytes, bandwidth and throughput

Network speed is measured in **bits per second**: Mbps (megabits) and Gbps (gigabits). File sizes are measured in **bytes**: MB and GB.

IMPORTANT

1 byte = 8 bits. A 100 Mbps connection can move at most about 12.5 megabytes per second.

- **Bandwidth** is the maximum a connection can carry, like the number of lanes on a road.
- **Throughput** is what you actually get, like the traffic that really gets through at rush hour. It is always lower than the bandwidth.

Key points

- Copper cables are cheap and common but limited to 100 m; fibre goes much further; Wi-Fi gives freedom but less speed and reliability.
- Network speeds are in bits per second; file sizes are in bytes. Divide by 8 to convert.
- Throughput is what you really get; bandwidth is the theoretical maximum.

HANDS-ON LAB · 2 min read

Lab: explore the network you are on

Use built-in commands to find your IP address, your gateway and your network card, and test your connection to the internet.

WHAT YOU WILL LEARN

- Open a command prompt or terminal
- Find your IP address, default gateway and MAC address
- Test a connection with ping
- See the path to a website with tracert / traceroute

What you need

- A Windows, macOS or Linux computer connected to a network (cable or Wi-Fi)
- About 20 minutes

NOTE

These commands only **read** information. They do not change any settings, so they are safe to try on any computer.

Step 1: open a command prompt

- **Windows:** press the Windows key, type `cmd` and press Enter.
- **macOS:** open **Terminal** from Applications → Utilities.
- **Linux:** open **Terminal**.

Step 2: find your network details

On Windows, type:

COMMAND PROMPT

```
ipconfig
```

Look for your active adapter (Ethernet or Wi-Fi) and write down:

Item	What it means
IPv4 Address	Your computer's address on this network, e.g. 192.168.1.23
Subnet Mask	Which part of the address is the network, e.g. 255.255.255.0
Default Gateway	The router your computer sends traffic to when it leaves this network

For more detail, including your network card's **physical (MAC) address** and your **DNS servers**, type:

COMMAND PROMPT

```
ipconfig /all
```

TIP

On macOS use `ifconfig` (or `ipconfig getifaddr en0` for just the address). On Linux use `ip addr` and `ip route` to see the gateway.

Step 3: test your connection with ping

`ping` sends a few small test messages and waits for replies.

COMMAND PROMPT

```
ping 192.168.1.1
```

Replace the address with **your** default gateway. Replies mean your computer can reach the router.

Now test the internet by address, then by name:

COMMAND PROMPT

```
ping 8.8.8.8  
ping google.com
```

- If `ping 8.8.8.8` works but `ping google.com` fails, the internet works but **name lookup (DNS)** does not.
- If neither works but the gateway replies, the problem is between your router and the internet provider.

NOTE

Some devices and websites are set not to answer ping, so no reply does not always mean a problem.

Step 4: see the path to a website

COMMAND PROMPT

```
tracert google.com
```

On macOS and Linux the command is `traceroute google.com`. Each line is a router your traffic passes through on its way. The first one is your own default gateway.

Check your understanding

1. What is your IPv4 address, and what is your default gateway?
2. What does it mean if you can ping 8.8.8.8 but not google.com?
3. How many routers ("hops") did your traffic pass through to reach google.com?

TIP

Save your answers. You will understand much more about each of these numbers after the **IP addressing** topic in the Foundations level.

ZERO · TYPES OF NETWORKS

NOTE · 2 min read

LAN, WAN, Wi-Fi and the internet: types of networks

Networks are named by the area they cover. Learn PAN, LAN, WLAN, MAN and WAN, and how the internet connects them all.

WHAT YOU WILL LEARN

- Describe PAN, LAN, WLAN, MAN and WAN with examples
- Explain what an internet service provider (ISP) does
- Explain why the internet is called a network of networks
- Tell the internet and an intranet apart

Networks by size

Networks are usually named by the **area they cover**.

Type	Stands for	Covers	Example
PAN	Personal Area Network	A few metres around one person	A phone connected to Bluetooth earphones
LAN	Local Area Network	One home, office or building	The computers and printers in an office
WLAN	Wireless LAN	The same as a LAN, but over Wi-Fi	The Wi-Fi in a café or school
MAN	Metropolitan Area Network	A town or city	A network linking a university's campuses across a city
WAN	Wide Area Network	Regions, countries, the world	A bank connecting its branches across Sierra Leone

NOTE

A LAN is usually **owned and managed by the organisation** that uses it. WAN links are usually **rented from a provider**, because they cross land and roads the organisation does not own.

The internet: a network of networks

The **internet** is not one network. It is millions of networks (homes, businesses, universities, mobile operators and data centres) connected to each other and all using the same protocols (TCP/IP).

Your connection to it comes from an **internet service provider (ISP)**: a mobile operator, a fixed-line or fibre provider, or a satellite service. ISPs connect to each other and to international cables, so your traffic can reach any other network in the world.

Internet, intranet and extranet

- **Internet:** the public, worldwide network of networks.
- **Intranet:** a private network for one organisation's staff only, such as an internal staff portal.
- **Extranet:** part of an intranet opened to selected outsiders, such as a supplier portal.

Key points

- LANs cover one site and are owned by the organisation; WANs connect sites far apart, usually over providers' links.
- A WLAN is simply a LAN that uses Wi-Fi.
- The internet is a worldwide network of networks; ISPs connect us to it.

CHEAT SHEET · 2 min read

Networking words every beginner should know

A one-page glossary of the words you will meet most often in this path, explained in plain language.

WHAT YOU WILL LEARN

- Recognise the most common networking terms
- Use this page as a quick reference while you learn

Glossary

Term	Plain-language meaning
Network	Devices connected to share information and resources
Node / host	Any device on a network, such as a laptop or printer
Client	A device or program that asks for a service
Server	A device or program that provides a service
Switch	Connects devices in the same network
Router	Connects different networks and chooses the path between them
Access point	Lets devices join a network over Wi-Fi
Firewall	Allows or blocks traffic according to rules
NIC	Network interface card: the part of a device that connects it to the network
MAC address	The hardware address of a network card, e.g. 3C-52-82-1A-7F-10
IP address	The logical address of a device on a network, e.g. 192.168.1.23
Subnet mask	Shows which part of an IP address identifies the network
Default gateway	The router a device sends traffic to when it leaves its own network
DNS	Turns names like google.com into IP addresses
DHCP	Gives devices their IP address and settings automatically
Protocol	An agreed set of rules for communicating
Bandwidth	The maximum amount of data a connection can carry
Throughput	The amount of data that actually gets through
Latency	The delay before data arrives, measured in milliseconds

Term	Plain-language meaning
Packet	A small piece of data sent across a network
ISP	Internet service provider: the company that connects you to the internet
LAN / WAN	A local network on one site / a network linking sites far apart
SSID	The name of a Wi-Fi network
Ping	A test that checks whether one device can reach another

TIP

Keep this page open in another tab while you study the other notes.

LEVEL 2 OF 5

Foundations

The core concepts every professional uses every day.

IP addressing & subnetting

How every device gets an address, and how networks are divided into smaller ones.

- What is an IP address?
- Binary for networking
- Subnetting step by step
- Subnetting cheat sheet
- Lab: subnet a small office network
- IPv6 essentials

The OSI and TCP/IP models

The layered models that explain how data travels from one application to another.

- The OSI model: seven layers explained
- The TCP/IP model and encapsulation
- TCP and UDP: reliable or fast
- Common ports and protocols

Ethernet and switching basics

How devices on the same network talk to each other: frames, MAC addresses and switches.

- Ethernet frames and MAC addresses
- How a switch learns and forwards
- ARP: finding a MAC address from an IP address
- Lab: build your first network in Packet Tracer

Essential network services

DHCP, DNS and NAT: the services that make networks easy to use.

- DHCP: automatic IP addressing
- DNS: turning names into addresses
- NAT and PAT: sharing one public address
- Lab: watch DHCP and DNS at work

FOUNDATIONS · IP ADDRESSING & SUBNETTING

NOTE · 3 min read

What is an IP address?

How IPv4 addresses are written, what the subnet mask and default gateway do, and the special addresses every technician should know.

WHAT YOU WILL LEARN

- Read and write an IPv4 address
- Explain what the subnet mask and default gateway are for
- Tell private and public addresses apart
- Recognise static, DHCP, loopback and APIPA addresses

An address for every device

Every device on a network needs an address so that information can be delivered to it, just as every house needs an address to receive letters. On today's networks this is the **IP address** (Internet Protocol address).

How an IPv4 address is written

An **IPv4 address** is 32 bits long. To make it readable, it is written as **four numbers from 0 to 255**, separated by dots. Each number is called an **octet** because it is 8 bits.

TEXT

192.168.1.23

Every address has two parts:

- the **network part**, shared by every device on the same network, and
- the **host part**, which is different for each device.

The subnet mask

The **subnet mask** shows where the network part ends and the host part begins.

Address	Subnet mask	Network part	Host part
192.168.1.23	255.255.255.0	192.168.1	23

With the mask 255.255.255.0 (also written **/24**), every device whose address starts with 192.168.1 is on the same network and can talk directly to the others.

The default gateway

When a device needs to reach an address **outside** its own network, for example a website, it sends the traffic to its **default gateway**: the router that connects the network to other networks. In a home or small office this is often 192.168.1.1 or 192.168.0.1.

Private and public addresses

Some address ranges are reserved for **private** use inside homes and organisations. They are not used on the internet itself.

Private range	Typically used by
10.0.0.0 – 10.255.255.255	Large organisations
172.16.0.0 – 172.31.255.255	Medium networks
192.168.0.0 – 192.168.255.255	Homes and small offices

Your router uses a **public** address (given by your ISP) on the internet side, and translates between it and your private addresses. This translation is called **NAT**, covered in the Essential network services topic.

How devices get their address

- **Dynamic (DHCP)**: the device asks for an address when it joins the network and is given one automatically. Most laptops and phones work this way.
- **Static**: the address is typed in by hand and never changes. Servers, printers and routers usually have static addresses so others can always find them.

Special addresses worth knowing

Address	Meaning
127.0.0.1	"Loopback": the device itself. Pinging it tests your own network software.
169.254.x.x	APIPA: the device asked for an address by DHCP and got no answer. A sign of a DHCP or cable problem.
0.0.0.0	"No address yet" or "any address", depending on where it appears

WARNING

Two devices must never share the same IP address on one network. If they do, both will have connection problems (an "IP address conflict").

What about IPv6?

The world is running out of IPv4 addresses, so a newer version, **IPv6**, uses 128-bit addresses written in hexadecimal, such as `2001:db8::1`. Most networks run both today. You will study IPv6 later in this path.

Key points

- An IPv4 address is four numbers from 0 to 255, with a network part and a host part.
- The subnet mask shows which part is which; the default gateway is the way out of the network.
- Private addresses are used inside networks; NAT connects them to the public internet.
- A 169.254.x.x address means the device could not reach a DHCP server.

NOTE · 2 min read

Binary for networking

The only maths you need for IP addressing: reading and writing 8-bit binary numbers, and how a subnet mask picks out the network.

WHAT YOU WILL LEARN

- Convert numbers between 0 and 255 to binary and back
- Write a subnet mask in binary and as a /prefix
- Find a network address by comparing an IP address with its mask

Why binary?

Computers store IP addresses as **bits**: ones and zeros. An IPv4 address is 32 bits, written as four groups of 8 bits (**octets**). To understand subnet masks and subnetting, you only need to work comfortably with 8-bit numbers.

The place values

Each of the 8 bits in an octet has a value. Reading from left to right:

Bit	1st	2nd	3rd	4th	5th	6th	7th	8th
Value	128	64	32	16	8	4	2	1

A bit set to 1 "switches on" its value. Add up the values that are on to get the decimal number.

Binary to decimal

TEXT

1 1 0 0 0 0 0 0	= 128 + 64	= 192
1 0 1 0 1 0 0 0	= 128 + 32 + 8	= 168
1 1 1 1 1 1 1 1	= 128+64+32+16+8+4+2+1	= 255
0 0 0 0 1 0 1 0	= 8 + 2	= 10

Decimal to binary

Work from the left: if the place value fits into what is left, write 1 and subtract it; if not, write 0.

Example: **172**

1. 128 fits → 1, leaves 44
2. 64 does not fit → 0
3. 32 fits → 1, leaves 12
4. 16 does not fit → 0

5. 8 fits → 1, leaves 4
6. 4 fits → 1, leaves 0
7. 2 → 0, 1 → 0

So **172 = 10101100**.

TIP

The numbers you see most in subnet masks are worth learning by heart: 128, 192, 224, 240, 248, 252, 254 and 255. Each one adds the next bit: 128, 128+64, 128+64+32, and so on.

Subnet masks in binary

A subnet mask is a row of 1s followed by a row of 0s. The **1s mark the network part** and the **0s mark the host part**. The number of 1s is the **prefix length**, written after a slash.

Mask	Binary	Prefix
255.255.255.0	11111111.11111111.11111111.00000000	/24
255.255.255.192	11111111.11111111.11111111.11000000	/26
255.255.0.0	11111111.11111111.00000000.00000000	/16

Finding the network address

A device finds its network by comparing its address with its mask bit by bit (a logical **AND**): where the mask has a 1, the address bit is kept; where the mask has a 0, the result is 0.

TEXT

Address	192.168.1.23	11000000.10101000.00000001.00010111
Mask	255.255.255.0	11111111.11111111.11111111.00000000
Network	192.168.1.0	11000000.10101000.00000001.00000000

In short: wherever the mask octet is 255, keep the address octet; wherever it is 0, write 0.

Practise

Convert to binary: **224, 10, 100**. Convert to decimal: **11110000, 00110010**.

Answers: 224 = 11100000, 10 = 00001010, 100 = 01100100, 11110000 = 240, 00110010 = 50.

NOTE · 3 min read

Subnetting step by step

Split one network into smaller ones: borrowing bits, working out the block size, and finding the network, hosts and broadcast of every subnet.

WHAT YOU WILL LEARN

- Explain why networks are divided into subnets
- Calculate the number of subnets and hosts for a prefix
- List the network address, usable hosts and broadcast address of each subnet
- Find which subnet an address belongs to

Why subnet?

Subnetting divides one network into several smaller ones. Organisations do it to:

- keep departments apart (for example Finance and Guest Wi-Fi), which improves **security**,
- reduce **broadcast traffic**, which every device on a network has to listen to, and
- use addresses **efficiently**, giving each group only what it needs.

Two formulas

When you make the prefix longer, you **borrow** host bits to create subnets.

- **Number of subnets** = 2 to the power of the bits borrowed
- **Usable hosts per subnet** = 2 to the power of the host bits left, **minus 2**

NOTE

You subtract 2 because, in every subnet, the first address is the **network address** and the last is the **broadcast address**. Neither can be given to a device.

Worked example: four subnets from a /24

An office has **192.168.10.0/24** and needs **4 networks**.

1. **Bits to borrow:** 2 bits give $2 \times 2 = 4$ subnets.
2. **New prefix:** $/24 + 2 = /26$, mask **255.255.255.192**.
3. **Host bits left:** $32 - 26 = 6$, so $2^6 - 2 = 62$ **usable hosts** per subnet.
4. **Block size:** $256 - 192 = 64$. Subnets start every 64 addresses.

Subnet	Network address	First usable	Last usable	Broadcast
1	192.168.10.0	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65	192.168.10.126	192.168.10.127

Subnet	Network address	First usable	Last usable	Broadcast
3	192.168.10.128	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193	192.168.10.254	192.168.10.255

TIP

The **block size** is the fastest shortcut in subnetting: 256 minus the interesting octet of the mask. Every subnet starts at a multiple of it.

Which subnet is an address in?

Question: which subnet does **192.168.10.150/26** belong to?

1. The block size for /26 is 64, so subnets start at 0, 64, 128 and 192.
2. 150 falls between 128 and 191.
3. So the network is **192.168.10.128**, the broadcast is **192.168.10.191** and the usable range is .129 to .190.

Choosing a prefix for a number of hosts

Work backwards from the hosts you need: find the smallest number of host bits whose $2^n - 2$ is big enough.

Hosts needed	Host bits	Prefix	Usable hosts
up to 6	3	/29	6
up to 14	4	/28	14
up to 30	5	/27	30
up to 62	6	/26	62
up to 126	7	/25	126

WARNING

Always leave room to grow. A department with 28 computers today will not fit in a /27 (30 hosts) once printers, phones and a few new staff arrive. Choose a /26.

Key points

- Borrowing bits creates subnets: 2^n subnets, each with $2^n - 2$ usable hosts.
- Block size = 256 – the interesting mask octet; subnets start at its multiples.
- The first address is the network, the last is the broadcast, and the ones between are usable.

CHEAT SHEET · 1 min read

Subnetting cheat sheet

Every common prefix with its mask, block size and number of usable hosts, plus the private ranges and special addresses.

WHAT YOU WILL LEARN

→ Look up the mask, block size and host count of any common prefix

Prefixes in the last octet

Prefix	Subnet mask	Block size	Usable hosts	Subnets in a /24
/24	255.255.255.0	256	254	1
/25	255.255.255.128	128	126	2
/26	255.255.255.192	64	62	4
/27	255.255.255.224	32	30	8
/28	255.255.255.240	16	14	16
/29	255.255.255.248	8	6	32
/30	255.255.255.252	4	2	64

Larger networks

Prefix	Subnet mask	Usable hosts
/16	255.255.0.0	65,534
/20	255.255.240.0	4,094
/21	255.255.248.0	2,046
/22	255.255.252.0	1,022
/23	255.255.254.0	510

Formulas

TEXT

Subnets = $2^{\text{(bits borrowed)}}$
Usable hosts = $2^{\text{(host bits)}} - 2$
Block size = $256 - (\text{interesting mask octet})$
Broadcast = next subnet's network address - 1

Private address ranges

Range	Prefix
10.0.0.0 – 10.255.255.255	10.0.0.0/8
172.16.0.0 – 172.31.255.255	172.16.0.0/12
192.168.0.0 – 192.168.255.255	192.168.0.0/16

Special addresses

Address	Meaning
127.0.0.1	Loopback (the device itself)
169.254.0.0/16	APIPA: no DHCP server answered
255.255.255.255	Broadcast to every device on the local network
0.0.0.0/0	The default route ("everywhere else")

TIP

Point-to-point links between two routers often use a **/30** (2 usable addresses), one for each end.

HANDS-ON LAB · 2 min read

Lab: subnet a small office network

Plan the addresses for an office with four networks of different sizes, then check your answers.

WHAT YOU WILL LEARN

- Choose the right prefix for each group of devices
- Allocate subnets without overlaps, largest first
- Write down the network, usable range and broadcast for each subnet

The scenario

A training centre has been given the network **192.168.20.0/24** and needs four separate networks:

Network	Devices needed
Administration	50
Finance	25
Training lab	12
Link between two routers	2

Because the networks are different sizes, each will get a different prefix. This is called **variable-length subnet masking (VLSM)**.

Your task

For each network, work out:

1. the prefix and subnet mask,
2. the network address,
3. the first and last usable addresses,
4. the broadcast address.

TIP

Always allocate the largest network first, starting at the beginning of the range. Each next subnet starts right after the previous one's broadcast address. This keeps the subnets from overlapping.

Use a sheet of paper or a spreadsheet. Try it before looking at the answers.

Check your answers

Step 1: choose a prefix for each

Network	Needs	Prefix	Mask	Usable hosts
Administration	50	/26	255.255.255.192	62
Finance	25	/27	255.255.255.224	30
Training lab	12	/28	255.255.255.240	14
Router link	2	/30	255.255.255.252	2

Step 2: allocate, largest first

Network	Network address	First usable	Last usable	Broadcast
Administration	192.168.20.0/26	192.168.20.1	192.168.20.62	192.168.20.63
Finance	192.168.20.64/27	192.168.20.65	192.168.20.94	192.168.20.95
Training lab	192.168.20.96/28	192.168.20.97	192.168.20.110	192.168.20.111
Router link	192.168.20.112/30	192.168.20.113	192.168.20.114	192.168.20.115

Addresses **192.168.20.116 to 192.168.20.255** are still free for future networks.

NOTE

It is common practice to give the **router** the first usable address of each subnet (for example 192.168.20.1 for Administration), so everyone knows where the default gateway is.

Going further

Finance expects to grow to 40 devices next year. What would you change? (Answer: Finance would need a /26. Allocate Administration 192.168.20.0/26, Finance 192.168.20.64/26, then the training lab at 192.168.20.128/28 and the link at 192.168.20.144/30.)

NOTE · 2 min read

IPv6 essentials

Why IPv6 exists, how its addresses are written and shortened, and the address types you will meet first.

WHAT YOU WILL LEARN

- Explain why IPv6 was created
- Write an IPv6 address in full and in its short form
- Recognise global unicast, link-local and loopback addresses
- Describe how devices can configure an IPv6 address automatically

Why IPv6?

IPv4 has about 4.3 billion addresses, and they ran out years ago. Tricks like NAT have stretched them, but the long-term answer is **IPv6**, which uses **128-bit** addresses: enough for every device on Earth many times over.

Most networks today run **both** IPv4 and IPv6 side by side (this is called **dual stack**).

How an IPv6 address is written

An IPv6 address is eight groups of four **hexadecimal** digits (0–9 and a–f), separated by colons:

TEXT

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

Shortening an address

Two rules make addresses shorter:

1. **Drop leading zeros** in any group: `0db8` becomes `db8`, `0042` becomes `42`.
2. **Replace one run of all-zero groups with `::`**, once per address.

TEXT

```
Full:      2001:0db8:0000:0000:0000:ff00:0042:8329
Rule 1:    2001:db8:0:0:0:ff00:42:8329
Rule 2:    2001:db8::ff00:42:8329
```

WARNING

`::` can appear only **once** in an address. Otherwise it would be impossible to know how many zero groups each one stands for.

Address types you will meet first

Type	Starts with	Used for
Global unicast	2000::/3 (usually 2 or 3)	Public addresses, reachable on the internet
Link-local	fe80::/10	Talking to devices on the same link only; every IPv6 interface has one
Unique local	fc00::/7 (in practice fd)	Private addresses inside an organisation
Loopback	::1	The device itself (like 127.0.0.1)

NOTE

IPv6 has **no broadcast addresses**. It uses **multicast** instead, which reaches only the devices that need a message.

Subnets in IPv6

In IPv6 the standard size for a network (a LAN) is a **/64**: the first 64 bits identify the network and the last 64 bits identify the device. Organisations typically receive a /48 or /56 from their provider and divide it into many /64 networks.

Getting an address automatically

IPv6 devices can configure their own address using **SLAAC** (stateless address autoconfiguration): the router announces the network's prefix, and the device creates the rest of the address itself. Networks can also use **DHCPv6**, much like DHCP in IPv4.

Try it

On Windows, run `ipconfig` and look for **Link-local IPv6 Address** (it starts with fe80). If your provider supports IPv6, you may also see an address starting with 2.

Key points

- IPv6 addresses are 128 bits, written as eight groups of hexadecimal digits.
- Drop leading zeros and use `::` once to shorten them.
- Every interface has a link-local fe80 address; global addresses start with 2 or 3; ::1 is loopback.

FOUNDATIONS · THE OSI AND TCP/IP MODELS

NOTE · 2 min read

The OSI model: seven layers explained

The seven-layer model network professionals use to describe how data moves and to locate problems.

WHAT YOU WILL LEARN

- Name the seven OSI layers in order
- Describe what each layer does with an everyday example
- Match devices and protocols to their layer
- Use the model to troubleshoot from the bottom up

Why a model?

Sending a message across a network involves many jobs: turning bits into signals, finding the right device, making sure nothing is lost, and presenting the result to an application. The **OSI model** (Open Systems Interconnection) splits these jobs into **seven layers**, each doing one part and relying on the layer below.

You will hear network engineers say things like "that's a layer 2 problem" or "a layer 3 switch". This is the model they mean.

The seven layers

#	Layer	What it does	Examples
7	Application	The network services applications use	HTTP, HTTPS, DNS, email (SMTP)
6	Presentation	Formats, encrypts and compresses data so both sides understand it	Encryption (TLS), character encoding, image formats
5	Session	Opens, manages and closes conversations between applications	Keeping a login session going
4	Transport	Delivers data between applications, reliably or quickly	TCP, UDP, port numbers
3	Network	Addresses and routes data between networks	IP addresses, routers
2	Data link	Delivers data between devices on the same network	MAC addresses, Ethernet, switches, Wi-Fi
1	Physical	Sends bits as signals over the medium	Cables, connectors, radio, light

TIP

To remember the layers from 1 to 7: **Please Do Not Throw Sausage Pizza Away** (Physical, Data link, Network, Transport, Session, Presentation, Application).

Data has a different name at each layer

Layer	Name of the data unit
4 Transport	Segment (TCP) or datagram (UDP)
3 Network	Packet
2 Data link	Frame
1 Physical	Bits

Devices and their layer

- **Cables, connectors, repeaters:** layer 1
- **Switches, Wi-Fi access points:** layer 2
- **Routers:** layer 3
- **Firewalls:** layers 3 and 4, and modern firewalls look up to layer 7

Using the model to troubleshoot

When something does not work, check the layers **from the bottom up**:

1. **Physical:** is the cable plugged in? Are the port lights on?
2. **Data link:** is the device connected to the right switch port or Wi-Fi network?
3. **Network:** does it have a valid IP address and gateway? Can it ping the gateway?
4. **Transport and above:** is the service running? Is a firewall blocking the port? Is the website's name resolving (DNS)?

NOTE

A large share of network problems turn out to be at layer 1: a loose cable, a damaged connector or a switched-off device. Always check the simple things first.

Key points

- The OSI model has seven layers, from Physical (1) to Application (7).
- Switches work mainly at layer 2 with MAC addresses; routers at layer 3 with IP addresses.
- Troubleshooting from layer 1 upwards saves time.

NOTE · 2 min read

The TCP/IP model and encapsulation

The four-layer model the internet actually runs on, and how each layer wraps the data in its own header on the way out.

WHAT YOU WILL LEARN

- Name the four layers of the TCP/IP model
- Map the TCP/IP layers to the OSI layers
- Explain encapsulation and de-encapsulation
- Describe what each header adds

The model the internet uses

The OSI model is a teaching and troubleshooting tool. The internet itself is built on the **TCP/IP model**, which groups the same jobs into **four layers**.

TCP/IP layer	Matches OSI layers	Main protocols
Application	7 Application, 6 Presentation, 5 Session	HTTP, HTTPS, DNS, DHCP, SMTP, SSH
Transport	4 Transport	TCP, UDP
Internet	3 Network	IPv4, IPv6, ICMP
Network access (link)	2 Data link, 1 Physical	Ethernet, Wi-Fi

Encapsulation: wrapping the data

When you send something, each layer adds its own **header** (and the link layer also a trailer) before passing the data down. This is called **encapsulation**, like putting a letter in an envelope, then the envelope in a parcel, then the parcel in a delivery van.

1. **Application:** your browser creates the request, for example "send me the home page of adramtechnologies.com".
2. **Transport:** TCP adds a header with the **source and destination port numbers** (for example 443 for HTTPS) and sequence numbers → a **segment**.
3. **Internet:** IP adds a header with the **source and destination IP addresses** → a **packet**.
4. **Network access:** Ethernet or Wi-Fi adds a header with the **source and destination MAC addresses** and a trailer with an error check → a **frame**, sent as **bits** on the medium.

TEXT

[Ethernet header | IP header | TCP header | HTTP request | Ethernet trailer]
MAC addresses IP addresses ports your data error check

De-encapsulation: unwrapping at the other end

The receiving device works in reverse: each layer reads its own header, checks it, removes it and passes the rest up, until the web server's software receives the original request.

NOTE

Routers along the way only unwrap up to layer 3: they read the IP header to decide where to send the packet, then build a **new frame** with new MAC addresses for the next link. The IP addresses stay the same from end to end; the MAC addresses change at every hop.

Key points

- TCP/IP has four layers: Application, Transport, Internet and Network access.
- Each layer adds a header on the way out (encapsulation) and removes it on the way in.
- Ports identify applications, IP addresses identify devices across networks, and MAC addresses identify devices on one link.

NOTE · 2 min read

TCP and UDP: reliable or fast

The two transport protocols, the three-way handshake, and how port numbers deliver data to the right application.

WHAT YOU WILL LEARN

- Compare TCP and UDP and give uses for each
- Describe the TCP three-way handshake
- Explain what a port number and a socket are
- Know the three ranges of port numbers

Two ways to deliver

At the transport layer, applications choose between two protocols.

	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection	Sets up a connection first	Just sends
Reliability	Checks that every segment arrives; resends lost ones	No checking or resending
Order	Puts data back in the right order	Data may arrive out of order
Speed	Slower, more overhead	Faster, very little overhead
Used for	Websites, email, file transfers, remote logins	Voice and video calls, live streaming, online games, DNS lookups

TIP

Ask: "Is it worse to be late or to be incomplete?" A web page or a file must arrive complete, so it uses **TCP**. In a voice call a late sound is useless, so it uses **UDP** and simply carries on.

The TCP three-way handshake

Before sending data, TCP opens a connection in three steps:

1. **SYN**: the client says "I'd like to talk" and sends its starting sequence number.
2. **SYN-ACK**: the server agrees and sends its own sequence number.
3. **ACK**: the client confirms. The connection is open.

Afterwards, each side **acknowledges** what it receives. If an acknowledgement does not come back in time, the sender resends the data. The connection is closed politely with FIN messages.

Port numbers

A device runs many applications at once, so the transport layer uses **port numbers** to deliver data to the right one. A web server listens on port **443** for HTTPS; your browser uses a random high port for its side of the conversation.

Range	Name	Used by
0 – 1023	Well-known ports	Standard services such as HTTP (80), HTTPS (443), SSH (22)
1024 – 49151	Registered ports	Specific applications, such as databases
49152 – 65535	Dynamic (private) ports	Chosen temporarily by clients for their side of a conversation

Sockets

An **IP address plus a port** is called a **socket**, for example `192.168.1.23:52100`. A conversation is identified by the pair of sockets at each end, which is how one computer can have dozens of browser tabs open to the same website without mixing them up.

Key points

- TCP is reliable and ordered; UDP is fast and simple.
- TCP opens connections with SYN, SYN-ACK, ACK.
- Ports identify applications; an IP address and a port together form a socket.

CHEAT SHEET · 2 min read

Common ports and protocols

The port numbers and protocols every network technician meets, with what they are for.

WHAT YOU WILL LEARN

→ Look up the port and transport protocol of common services

Ports to know by heart

Port	Protocol	Transport	Used for
20, 21	FTP	TCP	File transfer (unencrypted)
22	SSH / SFTP	TCP	Secure remote login and secure file transfer
23	Telnet	TCP	Remote login, unencrypted : avoid it
25	SMTP	TCP	Sending email between mail servers
53	DNS	UDP (and TCP)	Turning names into IP addresses
67, 68	DHCP	UDP	Giving out IP addresses (server 67, client 68)
80	HTTP	TCP	Websites (unencrypted)
110	POP3	TCP	Downloading email
123	NTP	UDP	Keeping clocks in time
143	IMAP	TCP	Reading email on the server
161, 162	SNMP	UDP	Monitoring network devices
443	HTTPS	TCP	Secure websites
465, 587	SMTP (secure)	TCP	Sending email from an email app
993	IMAPS	TCP	Reading email, encrypted
995	POP3S	TCP	Downloading email, encrypted
3306	MySQL	TCP	Database
3389	RDP	TCP	Windows Remote Desktop

WARNING

Never open **Remote Desktop (3389)**, **Telnet (23)** or database ports directly to the internet. They are among the most attacked ports. Use a VPN instead.

Protocols without ports

Some protocols work at layer 3 and have no port numbers:

Protocol	Used for
ICMP	Ping and traceroute, error messages
ARP	Finding a MAC address from an IPv4 address (layer 2)
OSPF, EIGRP	Routers sharing routes

FOUNDATIONS · ETHERNET AND SWITCHING BASICS

NOTE · 2 min read

Ethernet frames and MAC addresses

What is inside an Ethernet frame, how MAC addresses are built, and the difference between unicast, broadcast and multicast.

WHAT YOU WILL LEARN

- Describe the parts of an Ethernet frame
- Read a MAC address and find its manufacturer part
- Tell unicast, broadcast and multicast frames apart
- Explain what the MTU is

Ethernet

Ethernet is the set of standards almost every wired LAN uses. It defines the cables and signals (layer 1) and how data is packaged and addressed on the local network (layer 2). Wi-Fi works in a very similar way at layer 2.

The Ethernet frame

At layer 2, data travels in **frames**.

Field	Size	Purpose
Preamble	8 bytes	Lets the receiver lock on to the incoming signal
Destination MAC	6 bytes	Who the frame is for
Source MAC	6 bytes	Who sent it
Type	2 bytes	What is inside: 0x0800 = IPv4, 0x86DD = IPv6, 0x0806 = ARP
Data (payload)	46 – 1500 bytes	The packet being carried
FCS	4 bytes	An error check: damaged frames are thrown away

NOTE

The largest payload a standard Ethernet frame can carry is **1500 bytes**. This is called the **MTU** (maximum transmission unit). Bigger packets have to be split.

MAC addresses

Every network card has a **MAC address** (media access control address): 48 bits written as 12 hexadecimal digits.

TEXT

3C-52-82-1A-7F-10	(Windows style)
3c:52:82:1a:7f:10	(macOS and Linux style)
3c52.821a.7f10	(Cisco style)

- The **first half** (3C-52-82) is the **OUI**: it identifies the manufacturer.
- The **second half** is a number the manufacturer gives each card.

A MAC address only matters on the **local network**. It never leaves the LAN: routers replace the frame, and its MAC addresses, at every hop.

Three kinds of frame

Kind	Sent to	Example
Unicast	One device	Your laptop sending a page request to the router
Broadcast	Every device on the LAN (FF-FF-FF-FF-FF-FF)	An ARP request, a DHCP Discover
Multicast	A group of devices that asked to receive it	Some video streaming, routing protocol updates

TIP

Run `getmac` (Windows) or `ip link` (Linux) to see your own MAC addresses. Wi-Fi and Ethernet each have their own.

Key points

- Ethernet frames carry packets across one local network, from MAC address to MAC address.
- A MAC address is 48 bits; the first half identifies the manufacturer.
- Broadcasts reach every device on the LAN; unicast reaches one.

NOTE · 2 min read

How a switch learns and forwards

The MAC address table, flooding, and why switches replaced hubs: collision and broadcast domains made simple.

WHAT YOU WILL LEARN

- Explain how a switch builds its MAC address table
- Describe what a switch does with known, unknown and broadcast frames
- Define collision domains and broadcast domains
- Explain why switches replaced hubs

The switch's job

A **switch** connects the devices of a LAN and delivers each frame **only to the port where its destination lives**. To do that, it keeps a list of which MAC address is on which port: the **MAC address table**.

Learning

Every time a frame arrives, the switch looks at its **source MAC address** and writes down the port it came in on.

MAC address	Port
3c52.821a.7f10	Fa0/1
00d0.97ab.1c22	Fa0/2
0060.2f84.9a01	Fa0/5

Entries that are not seen again are removed after a while (on Cisco switches, after **5 minutes** by default), so the table stays up to date when devices move.

Forwarding

Then the switch looks at the **destination MAC address**:

The destination is...	The switch...
In the table	Sends the frame out of that one port only (forwarding)
Not in the table	Sends it out of every port except the one it came in on (flooding)
The broadcast address	Floods it to every port except the incoming one
On the same port it came from	Drops it (filtering)

As soon as the unknown device replies, the switch learns its port, and from then on frames go straight to it.

Collision and broadcast domains

- A **collision domain** is a part of the network where two devices sending at the same moment would interfere. Old **hubs** repeated every signal to every port, so the whole hub was one collision domain. On a switch, **each port is its own collision domain**, and with **full-duplex** links devices can send and receive at the same time without collisions.
- A **broadcast domain** is the set of devices that receive each other's broadcasts. A switch forwards broadcasts everywhere, so **all its ports are one broadcast domain**. **Routers stop broadcasts**: each router interface is a separate broadcast domain. (VLANs, in the Intermediate level, also split a switch into several broadcast domains.)

NOTE

Hubs are obsolete. If you find one in an old office, replace it with a switch: performance and security both improve.

Seeing the table

On a Cisco switch:

CISCO IOS

```
Switch> enable
Switch# show mac address-table
```

Key points

- Switches learn source MAC addresses and forward by destination MAC address.
- Unknown and broadcast frames are flooded; known ones go to one port.
- Each switch port is a collision domain; the whole switch is one broadcast domain until VLANs or routers split it.

NOTE · 2 min read

ARP: finding a MAC address from an IP address

How a device discovers the MAC address it needs before it can send a frame, and why traffic to other networks goes to the gateway's MAC.

WHAT YOU WILL LEARN

- Explain why ARP is needed
- Describe an ARP request and an ARP reply
- Explain which MAC address is used to reach a device on another network
- View and read the ARP cache

The missing piece

To send a packet on the local network, a device must put a **destination MAC address** in the frame. But applications only know **IP addresses**. The **Address Resolution Protocol (ARP)** links the two for IPv4.

How ARP works

Say 192.168.1.10 wants to send to 192.168.1.20 on the same network:

1. **ARP request (broadcast):** "Who has 192.168.1.20? Tell 192.168.1.10." Every device on the LAN receives it.
2. **ARP reply (unicast):** only 192.168.1.20 answers: "192.168.1.20 is at 00d0.97ab.1c22."
3. The sender saves the answer in its **ARP cache** and can now build the frame.

The cache entry is kept for a short time, so ARP does not have to ask again for every packet.

Sending to another network

If the destination is on a **different network** (for example a website on the internet), the device does **not** ARP for the website's address. It knows, thanks to its subnet mask, that the destination is outside its network, so it:

1. ARPs for its **default gateway's** MAC address, and
2. sends the frame to the router, with the website's **IP address** still in the packet.

The router then forwards the packet towards the destination, building a new frame at each hop.

TIP

If a computer can talk to devices on its own network but not to anything outside, check its **default gateway** setting. With a wrong gateway, it ARPs for a router that does not exist.

Looking at the ARP cache

COMMAND PROMPT

```
arp -a
```

Each line shows an IP address, the MAC address it belongs to, and whether it was learned automatically (dynamic) or set by hand (static).

WARNING

ARP trusts any reply it receives. Attackers can abuse this with **ARP spoofing**, pretending to be the gateway so traffic passes through their computer. Managed switches offer protections such as dynamic ARP inspection, covered in the security topics.

Key points

- ARP finds the MAC address that belongs to an IPv4 address on the local network.
- Requests are broadcast; replies are unicast; answers are cached.
- To reach another network, a device sends frames to its default gateway's MAC address.

HANDS-ON LAB · 2 min read

Lab: build your first network in Packet Tracer

Connect two computers through a switch in Cisco Packet Tracer, give them addresses, test with ping and look at the switch's MAC address table.

WHAT YOU WILL LEARN

- Add and cable devices in Packet Tracer
- Configure static IP addresses on PCs
- Test connectivity with ping
- Read a switch's MAC address table

What you need

- **Cisco Packet Tracer**, a free network simulator. Download it from Cisco Networking Academy ([netacad.com](https://www.netacad.com)) after creating a free account.
- About 30 minutes

Step 1: add the devices

1. Open Packet Tracer and start a new file.
2. From **Network Devices** → **Switches**, drag a **2960** switch onto the workspace.
3. From **End Devices**, drag two **PCs** next to it. Name them **PC1** and **PC2** (click a device, then the **Config** tab).

Step 2: connect them

1. Choose **Connections**, then the **Copper Straight-Through** cable (solid black line).
2. Click PC1, choose **FastEthernet0**, then click the switch and choose **FastEthernet0/1**.
3. Connect PC2's **FastEthernet0** to the switch's **FastEthernet0/2**.

The link lights turn from orange to green after a few seconds while the switch ports come up.

NOTE

Straight-through cables connect different kinds of devices (PC to switch). Modern equipment detects the cable type automatically, but knowing the difference still helps in exams and older networks.

Step 3: give each PC an address

Click PC1 → **Desktop** → **IP Configuration** → **Static**, and enter:

PC	IPv4 address	Subnet mask
PC1	192.168.10.10	255.255.255.0
PC2	192.168.10.11	255.255.255.0

No default gateway is needed: both PCs are on the same network.

Step 4: test with ping

On PC1, open **Desktop** → **Command Prompt** and type:

```
COMMAND PROMPT
```

```
ping 192.168.10.11
```

You should see four replies. The first one may time out while ARP finds PC2's MAC address. That is normal.

Step 5: look inside the switch

Click the switch → **CLI** tab, press Enter, and type:

```
CISCO IOS
```

```
enable  
show mac address-table
```

You will see both PCs' MAC addresses, each next to the port it is connected to. The switch learned them from the ping traffic.

TIP

Switch to **Simulation** mode (bottom right) and ping again. You can watch each frame travel, including the ARP broadcast reaching every port and the reply going to one port only.

Check your understanding

1. Why did the first ping sometimes fail?
2. Change PC2's mask to 255.255.0.0. Does ping still work? Why?
3. Add a third PC on port Fa0/3 with the address 192.168.20.12/24. Can PC1 reach it? Why not?

(For question 3: it is on a different network, and there is no router to connect the two.)

FOUNDATIONS · ESSENTIAL NETWORK SERVICES

NOTE · 2 min read

DHCP: automatic IP addressing

How devices get their IP address, mask, gateway and DNS servers automatically, the four DHCP messages, and what scopes, leases and reservations are.

WHAT YOU WILL LEARN

- Explain what DHCP gives a device
- Describe the four DHCP messages (DORA)
- Explain leases, scopes, exclusions and reservations
- Recognise the signs of a DHCP problem

Why DHCP?

Typing an IP address, mask, gateway and DNS server into every laptop and phone by hand would be slow and full of mistakes. The **Dynamic Host Configuration Protocol (DHCP)** hands these settings out automatically.

A DHCP server usually gives each device:

- an **IP address** and **subnet mask**
- a **default gateway**
- one or more **DNS servers**
- a **lease time**: how long it may keep the address

In a home or small office the router is the DHCP server. Larger organisations often use a server (for example Windows Server) or the main router or firewall.

The four messages: DORA

Step	Message	Sent by	What it means
1	Discover	Client (broadcast)	"Is there a DHCP server here?"
2	Offer	Server	"You can have 192.168.1.50, with these settings."
3	Request	Client (broadcast)	"I'd like 192.168.1.50, please."
4	Acknowledge	Server	"It's yours for 24 hours."

The client starts with no address, so its first messages are **broadcasts**, using UDP port 68 (client) and 67 (server).

Scopes, exclusions, leases and reservations

- **Scope (pool):** the range of addresses the server may hand out, for example 192.168.1.100 to 192.168.1.200.
- **Exclusions:** addresses inside the range the server must not give out, because they are used by printers or servers with static addresses.
- **Lease:** a device keeps its address for the lease time and **renews** it halfway through. Addresses from devices that leave are reused later.
- **Reservation:** the server always gives the **same address** to a particular device, recognised by its MAC address. Useful for printers.

TIP

Keep static addresses (routers, servers, printers) **outside** the DHCP scope, or exclude them. Otherwise DHCP may hand the same address to a laptop and cause a conflict.

DHCP across routers

Broadcasts do not cross routers. When the DHCP server is on a different network, the router is configured as a **DHCP relay** (on Cisco: `ip helper-address`) to forward the requests to it.

Signs of a DHCP problem

- The device shows an address starting with **169.254** (APIPA): it got no answer from any DHCP server.
- "IP address conflict" messages: check for static addresses inside the scope.
- New devices connect but older ones do not, or the other way round: the scope may be **full**.

Key points

- DHCP hands out the IP address, mask, gateway and DNS servers automatically.
- DORA: Discover, Offer, Request, Acknowledge.
- Use exclusions for static devices, reservations for devices that need a fixed address, and relays across routers.

NOTE · 2 min read

DNS: turning names into addresses

How a name like `adramtechnologies.com` becomes an IP address, the DNS hierarchy, caching, and the record types you will meet.

WHAT YOU WILL LEARN

- Explain why DNS is needed
- Describe how a DNS lookup travels from resolver to root, TLD and authoritative servers
- Recognise A, AAAA, CNAME, MX and other common records
- Use `nslookup` to test DNS

Names for people, numbers for computers

People remember names such as **google.com**; networks deliver to **IP addresses**. The **Domain Name System (DNS)** is the internet's directory that turns one into the other.

The DNS hierarchy

DNS is spread across many servers, organised like a tree:

Level	Example	Knows...
Root servers	.	Where to find each top-level domain
Top-level domain (TLD)	.com, .org, .sl	Which servers are responsible for each domain under it
Authoritative servers	The servers for example.com	The actual records for that domain

How a lookup works

When you type a website name:

1. Your computer checks its own **cache**. If it looked the name up recently, it already knows the answer.
2. If not, it asks its configured DNS server, the **recursive resolver** (often the router, the ISP's server, or a public one such as 8.8.8.8 or 1.1.1.1).
3. The resolver asks a **root server**, which points it to the **.com** servers.
4. The **.com** servers point it to the domain's **authoritative** servers.
5. The authoritative server gives the answer, for example 93.184.215.14.
6. The resolver returns the answer to your computer, and both **cache** it for the time set in the record (its **TTL**).

DNS lookups normally use **UDP port 53** (TCP for large answers).

Common record types

Record	Holds	Example
A	An IPv4 address	www → 203.0.113.10
AAAA	An IPv6 address	www → 2001:db8::10
CNAME	Another name (an alias)	shop → www.example.com
MX	The mail servers for the domain	example.com → mail.example.com
NS	The domain's authoritative name servers	example.com → ns1.host.com
TXT	Text, often for email security and verification (SPF, DKIM)	"v=spf1 ..."
PTR	The name for an IP address (reverse lookup)	203.0.113.10 → www.example.com

Testing DNS with nslookup

COMMAND PROMPT

```
nslookup example.com
nslookup -type=mx gmail.com
```

The first command shows the IP address of example.com; the second shows Gmail's mail servers.

TIP

If websites fail to open by name but `ping 8.8.8.8` works, the problem is DNS. Try another DNS server, or clear the cache with `ipconfig /flushdns` on Windows.

Key points

- DNS turns names into IP addresses using a hierarchy of root, TLD and authoritative servers.
- Answers are cached for their TTL, so most lookups are instant.
- A records hold IPv4 addresses, AAAA IPv6, CNAME aliases and MX mail servers.

NOTE · 2 min read

NAT and PAT: sharing one public address

How a router lets a whole office share one public IP address, the three kinds of NAT, and what port forwarding does.

WHAT YOU WILL LEARN

- Explain why NAT is needed
- Compare static NAT, dynamic NAT and PAT
- Read a simple NAT translation table
- Explain port forwarding and carrier-grade NAT

Why NAT?

Private addresses (10.x, 172.16–31.x, 192.168.x) cannot be used on the internet. **Network Address Translation (NAT)** lets a router swap a private address for a **public** one as traffic leaves the network, and swap it back for the replies. It also saves public IPv4 addresses, which are scarce.

Three kinds of NAT

Kind	How it works	Typical use
Static NAT	One private address is always mapped to one public address	A server inside the network that must be reachable from outside
Dynamic NAT	Private addresses borrow public ones from a pool, first come first served	Rare today
PAT (NAT overload)	Many private addresses share one public address, told apart by port numbers	Almost every home and office router

PAT in action

Your office has one public address, 203.0.113.5. Two staff open websites at the same time:

Inside (private)	Outside (public)	Website
192.168.1.20 : 51000	203.0.113.5 : 51000	142.250.1.1 : 443
192.168.1.35 : 51000	203.0.113.5 : 51001	142.250.1.1 : 443

Both share 203.0.113.5. The router changes the **port** when two devices pick the same one, and remembers each mapping in its **translation table** so replies go back to the right computer.

Port forwarding

PAT only lets **outgoing** conversations start. To make a device inside reachable from the internet (for example a CCTV recorder or a web server), you add a **port forward**: "traffic arriving on public port 8443 goes to 192.168.1.50 port 443".

WARNING

Every port forward opens a door from the internet into your network. Only forward what is truly needed, keep the device updated, and prefer a **VPN** for remote access to cameras and computers.

Carrier-grade NAT

Many mobile and some fixed internet providers put **many customers behind a shared public address** using another layer of NAT (carrier-grade NAT, CGNAT). On such connections port forwarding usually cannot work, because your router does not have its own public address. A business that needs to host services should ask its provider for a **static public IP address**.

Key points

- NAT translates private addresses to public ones at the edge of the network.
- PAT lets a whole network share one public address using port numbers.
- Port forwarding makes an inside device reachable from outside: use it sparingly.

HANDS-ON LAB · 2 min read

Lab: watch DHCP and DNS at work

Use built-in commands to see your DHCP lease, renew it, look up names with nslookup and inspect your DNS cache.

WHAT YOU WILL LEARN

- Find the DHCP server and lease times for your computer
- Release and renew a DHCP address
- Look up A and MX records with nslookup
- View and clear the DNS cache

What you need

- A Windows computer connected to a network that uses DHCP (almost all home and office networks)
- About 20 minutes

WARNING

Step 2 disconnects your computer from the network for a few seconds. Do not do it while connected to the computer remotely, or during an online exam or call.

Step 1: read your DHCP lease

COMMAND PROMPT

```
ipconfig /all
```

For your active adapter, find:

Line	What it tells you
DHCP Enabled	Yes means the address came from DHCP
DHCP Server	The device that gave you the address (often your router)
Lease Obtained / Lease Expires	When you got the address and when it must be renewed
DNS Servers	Where your computer sends its DNS lookups

Step 2: release and renew

COMMAND PROMPT

```
ipconfig /release  
ipconfig /renew
```

After `/release` the adapter has no IPv4 address. `/renew` runs the full **DORA** exchange: Discover, Offer, Request, Acknowledge. Run `ipconfig /all` again: the **Lease Obtained** time has changed.

Step 3: look up names

COMMAND PROMPT

```
nslookup example.com
nslookup -type=mx gmail.com
nslookup example.com 1.1.1.1
```

- The first shows the **A record** (IPv4 address) of example.com, and which server answered.
- The second shows Gmail's **MX records**: its mail servers.
- The third asks a specific DNS server, **1.1.1.1**, instead of your usual one.

Step 4: the DNS cache

COMMAND PROMPT

```
ipconfig /displaydns
ipconfig /flushdns
```

The first lists the names your computer has looked up recently and their TTLs. The second empties the cache, so the next lookup goes to the DNS server again.

TIP

On macOS: `ipconfig getpacket en0` shows the DHCP lease, and `nslookup` works the same way. On Linux, `dig example.com` gives more detail than nslookup.

Check your understanding

1. What is the address of your DHCP server, and how long is your lease?
2. Which DNS servers does your computer use? Are they your router, your provider or a public service?
3. What address did you get for example.com from your usual server and from 1.1.1.1? Were they the same?